

## 1. Signals – Classification and Basic Signals

### ❖ Classification of Signals

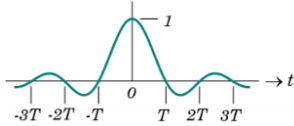
- Periodic Signals:  $x(t) = x(t + T); \forall t$
- Complex Signals:  $z = re^{j\theta} = r\cos(\theta) + j\sin(\theta)$
- Energy Signals

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt, 0 < E < \infty$$

- Power Signals

$$P = \lim_{\tau \rightarrow \infty} \frac{1}{2\tau} \int_{-\tau}^{\tau} |x(t)|^2 dt, 0 < P < \infty$$

### ❖ Basic Signals

- $\text{sinc}\left(\frac{t}{T}\right) = \begin{cases} \frac{\sin(\pi t/T)}{\pi t/T}; & t \neq 0 \\ 1; & t = 0 \end{cases}$  
- $x(t) = \mu e^{j(\omega_0 t + \phi)} = \mu[\cos(\omega_0 t + \phi) + j\sin(\omega_0 t + \phi)]$

## Time Domain

## 2. Time-domain Operations & the Dirac Impulse

### ❖ Time-domain Operations

- Time-scaling; Sum; Product; Convolution

### ❖ The Dirac- $\delta$ function (Unit Impulse function) $\delta(t)$

$$\delta(t) = \begin{cases} \infty; & t = 0 \\ 0; & t \neq 0 \end{cases} \text{ and } \int_{-\varepsilon}^{\varepsilon} \delta(t) dt = 1, \forall \varepsilon > 0$$

- Properties of  $\delta(t)$

- Symmetry:  $\delta(t) = \delta(-t)$
- Sampling:  $x(t)\delta(t - \lambda) = x(\lambda)\delta(t - \lambda)$
- Sifting:  $\int_{-\infty}^{\infty} x(t)\delta(t - \lambda) dt = x(\lambda)$
- Replication:  $x(t) * \delta(t - \lambda) = x(t - \lambda)$

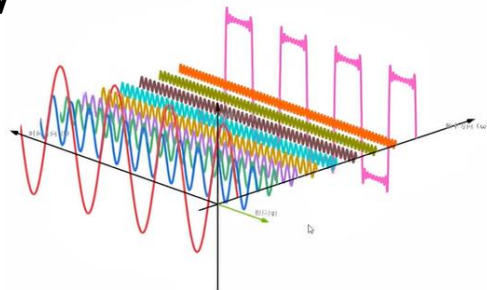
- The Dirac- $\delta$  Comb function:

$$\text{comb}_{\lambda}(t) = \sum_n \delta(t - n\lambda)$$

Model a periodic signal:  $x_p(t) = x(t) * \sum_n \delta(t - n\lambda)$

Model a sampled signal:  $x_s(t) = x(t) \times \sum_n \delta(t - n\lambda)$

## Frequency Domain



$$\begin{aligned} \tilde{x}(t) &= \mu e^{j(2\pi f_0 t + \phi)} \\ &= \underbrace{\mu e^{j\phi}}_{\text{Spectrum}} \times \underbrace{e^{j2\pi f_0 t}}_{\text{Normalized complex sinusoid}} \end{aligned}$$

## 5. Spectral Density and Bandwidth

### ❖ Rayleigh Energy Theorem

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$$

### ❖ Energy Spectral Density

$$E_x(f) = |X(f)|^2$$

### ❖ Parseval Power Theorem

$$\begin{aligned} P &= \lim_{W \rightarrow \infty} \frac{1}{2W} \int_{-W}^W |x(t)|^2 dt \\ &= \int_{-\infty}^{\infty} \lim_{W \rightarrow \infty} \frac{1}{2W} |X(f)|^2 df = \int_{-\infty}^{\infty} P_x(f) df \end{aligned}$$

For periodic signals:

$$P_x(f) = \sum_{k=-\infty}^{\infty} |c_k|^2 \delta(f - kf_p)$$

$$P = \int_{-\infty}^{\infty} P_x(f) df = \sum_{k=-\infty}^{\infty} |c_k|^2$$

## 3. Discrete-Frequency Spectrum (Fourier Series)

An expansion of a periodic function into a sum of sinusoids

### ❖ Complex Exponential Fourier Series

- Fourier series coefficients

$$c_k = \frac{1}{T_p} \int_{t_0}^{t_0+T_p} x_p(t) e^{-j2\pi kt/T_p} dt, k = 0, \pm 1, \pm 2, \dots$$

- Fourier series expansion

$$x_p(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi kt/T_p}$$

## 4. Continuous-Frequency Spectrum (Fourier Transform)

The spectrum of an aperiodic signal in the continuous-frequency domain,  $f$ .

### ❖ Fourier Transform:

- Forward Fourier Transform:

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

- Inverse Fourier Transform:

$$x(t) = \int_{-\infty}^{+\infty} X(f) e^{j2\pi ft} df$$

### ❖ Properties of Fourier Transform:

- **Linearity; Time Scaling; Duality; Time Shifting; Frequency Shifting; Differentiation; Integration; Convolution.**

### ❖ Spectral Properties of a REAL Signal

### ❖ Spectrum of Signals that are not Absolutely Integrable

- Impulse:  $\mathfrak{F}\{K\delta(t)\} = K$
- DC:  $\mathfrak{F}\{K\} = K\delta(f)$
- Complex Exponential:  $\mathfrak{F}\{K e^{j2\pi f_0 t}\} = K\delta(f - f_0)$
- Arbitrary periodic signals:

$$X_p(f) = \sum_{k=-\infty}^{\infty} c_k \delta\left(f - \frac{k}{T_p}\right)$$

- Dirac- $\delta$  Comb Function:

$$\mathfrak{F}\{\text{comb}_{\lambda}(t)\} = \frac{1}{\lambda} \sum_k c_k \delta(f - k/\lambda)$$